SC2006 LAB 4

Software Testing Report

Pilot-aligned black-box and white-box evidence

Work product	Test cases, control-flow graphs and executed results
Prepared by	Ziru Gao — Group 1, Software Testing
System	Group Scheduling Application
Production baseline	feature/production-pilot @ 26cb622
SRS oracle	SRS v2.3 - Pilot near-match recovery
Execution date	7 November 2026 (Asia/Singapore)
Assessment result	40/40 assessed tests passed
Total Lab 4 evidence	48/48 tests passed (40 assessed + 8 supplemental)

Submission scope

This report satisfies Lab 4 Manual sections 3.2.1-3.2.4: one important control boundary is tested with equivalence classes and boundary values; two complex class methods are tested with basis paths; redundant cases are minimised; and every executed case records input, expected output, actual output and pass/fail status.

1 Scope, Requirement Audit and Baseline

Manual requirement	Evidence	Status
3.2.1 one control class	RecommendationEngine.generate() boundary; Sections 2-3	Met
3.2.1 equivalence classes	Seven input characteristics; Section 2.1	Met
3.2.1 boundary values	Duration, threshold, journey and weight neighbours	Met
3.2.2 two complex methods	AvailabilityService._validate_intervals and _ParticipantAvailability.status_for	Met
3.2.2 basis-path testing	Two CFGs; V(G)=6 and V(G)=9; 15 assessed paths	Met
3.2.3 minimise cases	One representative per partition plus necessary boundary neighbours	Met
3.2.4 execute/document	48-test JUnit; every assessment row has expected/actual/result	Met

1.1 Authoritative baseline

The tested product baseline is **feature/production-pilot** at commit **26cb622a8724018f7582225c640d5ac93e9ac8a5**. SRS v2.3 is the approved as-built requirements oracle for implemented Pilot behaviour. The assessment does not treat an older SRS contradiction as a reason to change working production behaviour.

Evidence layer	Version / result
Production code	Pilot commit 26cb622
Requirements	SRS v2.3: normal threshold, labelled recovery, explicit no-result
Assessment tests	25 black-box + 6 interval paths + 9 attendance paths = 40
Supplemental guards	7 pre-routing paths + 1 production-step guard
Fresh Lab 4 run	48 passed; 0 failed; 0 errors; 0 skips

Product commit 26cb622 · Execution-evidence commit 57ee158

No production method was changed to make this report pass; only SRS documentation, tests, CFGs and this report were added or strengthened.

2 Black-box Test Design

The recommendation engine is treated as a sealed control boundary. Tests construct complete RecommendationRequest values and assert only returned RecommendationResult data or public validation errors. The deterministic router replaces only the external network provider; candidate generation, feasibility, routing adaptation, scoring and ranking remain real production code.

2.1 Equivalence classes

Characteristic	Equivalence classes	Cases
Duration	invalid <15; valid 15-480; invalid >480	BB-DUR-01..06
Requested attendance	met; missed with best>=2; best<2; invalid 0; explicit 1	BB-THR-01..05
Venue support	none; mixed catalogue with one supporting venue	BB-VEN-01..02
Starting locations	valid centroid; no valid active origins	BB-THR-01, BB-LOC-01
Routing result	available; unavailable; exception; malformed available-without-duration	BB-JRN-01, BB-RTE-01..03
Journey duration	at cap; above cap	BB-JRN-01..02
Attendance weight	invalid <0; valid 0-100 in 10% steps; invalid >100	BB-WGT-01..06

2.2 Boundary-value analysis

Boundary	Selected values	Expected relation	Cases
Duration minimum 15	14, 15, 16	reject / accept / accept	BB-DUR-01,02,05
Duration maximum 480	479, 480, 481	accept / accept / reject	BB-DUR-06,03,04
Threshold positivity	0, 1	reject / accept	BB-THR-04,05
Journey cap 60	60, 61	retain / exclude	BB-JRN-01,02
Weight minimum 0	-1, 0, 10	reject / accept / accept	BB-WGT-03,01,05
Weight maximum 100	90, 100, 101	accept / accept / reject	BB-WGT-06,02,04

2.3 Minimisation

Each non-boundary partition has one representative. Numeric boundaries use invalid/endpoint/valid-neighbour values where they distinguish a realistic defect. Shared valid setup is reused rather than multiplying unrelated combinations. The 25 cases therefore improve fault detection without a Cartesian-product explosion.

3 Black-box Cases and Results (1 of 3)

Case	Req.	Input / precondition	Expected	Actual	Result
BB-DUR-01	FR-A1.1	duration 14	Reject below 15	ValueError: duration must be 15 to 480 minutes	PASS
BB-DUR-02	FR-A1.1	duration 15	Accept lower boundary	OK; ranked options present	PASS
BB-DUR-03	FR-A1.1	duration 480	Accept upper boundary	OK; ranked options present	PASS
BB-DUR-04	FR-A1.1	duration 481	Reject above 480	ValueError: duration must be 15 to 480 minutes	PASS
BB-DUR-05	FR-A1.1	duration 16	Accept valid neighbour	OK; all generated slots are 16 minutes	PASS
BB-DUR-06	FR-A1.1	duration 479	Accept valid neighbour	OK; all generated slots are 479 minutes	PASS
BB-THR-01	FR-D1	2 available; requested 2	Normal result	OK; all ranked attendee_count=2	PASS
BB-THR-02	FR-D1.1 / FR-E1.1	best 2; requested 3	Label near match; preserve 3	BEST_AVAILABLE_BELOW_THRESHOLD; threshold 3; actual 2; warning names both	PASS
BB-THR-03	FR-D1.1	best 1; requested 2	No group result; no routing	NO_TIMESLOT_MEETS_THRESHOLD; 0 combinations; 0 route calls	PASS

Actual outputs above are literal conditions asserted by the named automated test; they are not copied from the expected-output column.

Mutation check

Representative defect	Test that would fail
Accept a 14-minute Session or alter a generated 16-minute slot	BB-DUR-01 / BB-DUR-05
Replace the preserved requested threshold 3 with another value	BB-THR-02
Route even though no two-person group can form	BB-THR-03

3 Black-box Cases and Results (2 of 3)

Case	Req.	Input / precondition	Expected	Actual	Result
BB-THR-04	FR-A4	threshold 0	Reject non-positive value	ValueError: Minimum-Attendee Threshold must be positive	PASS
BB-THR-05	FR-A4 / FR-D1	1 available; requested 1	Accept explicit positive value	OK; threshold 1; ranked attendee_count=1	PASS
BB-VEN-01	FR-C3	badminton; football-only venue	No supporting venue	NO_SUPPORTING_VENUES; no ranked options	PASS
BB-VEN-02	FR-C3 / FR-C4	mixed supporting/non-supporting	Filter before shortlist	Only badminton-hall appears in shortlist and ranking	PASS
BB-LOC-01	FR-C4 / NFR-3	all origins invalid/unavailable	No centroid; no routing	NO_VALID_LOCATIONS; 0 route calls	PASS
BB-RTE-01	NFR-3	UNAVAILABLE; minutes None	Never invent duration	DATA_UNAVAILABLE; minutes None; no rank	PASS
BB-RTE-02	NFR-3	router raises RuntimeError	Contain provider error	Travel ERROR; DATA_UNAVAILABLE; no rank	PASS
BB-RTE-03	NFR-3	AVAILABLE but minutes None	Treat malformed success as error	Travel ERROR; minutes None; DATA_UNAVAILABLE	PASS

Actual outputs above are literal conditions asserted by the named automated test; they are not copied from the expected-output column.

Mutation check

Representative defect	Test that would fail
Let an unsupported football venue enter a badminton shortlist	BB-VEN-02
Convert a missing route duration into a fabricated zero	BB-RTE-01 / BB-RTE-03
Propagate a provider exception instead of containing it	BB-RTE-02

3 Black-box Cases and Results (3 of 3)

Case	Req.	Input / precondition	Expected	Actual	Result
BB-JRN-01	FR-D3	journey 60; cap 60	Keep exact boundary	OK; ranked; exceeded flag False	PASS
BB-JRN-02	FR-D3	journey 61; cap 60	Exclude above cap	No rank; exclusion JOURNEY_LIMIT	PASS
BB-WGT-01	FR-D9	weight 0%	Accept lower boundary	OK; reported weight 0; ranked	PASS
BB-WGT-02	FR-D9	weight 100%	Accept upper boundary	OK; reported weight 100; ranked	PASS
BB-WGT-03	FR-D9	weight -1%	Reject below 0	ValueError: attendance weight must be between 0 and 100	PASS
BB-WGT-04	FR-D9	weight 101%	Reject above 100	ValueError: attendance weight must be between 0 and 100	PASS
BB-WGT-05	FR-D9.1	weight 10%	Accept discrete neighbour	OK; reported weight 10; ranked	PASS
BB-WGT-06	FR-D9.1	weight 90%	Accept discrete neighbour	OK; reported weight 90; ranked	PASS

Actual outputs above are literal conditions asserted by the named automated test; they are not copied from the expected-output column.

Mutation check

Representative defect	Test that would fail
Exclude the exact 60-minute journey boundary	BB-JRN-01
Accept an attendance weight outside 0-100	BB-WGT-03 / BB-WGT-04
Snap the valid 10% or 90% neighbour to another value	BB-WGT-05 / BB-WGT-06

Near-match trace

BB-THR-02 specifically proves the SRS v2.3 recovery contract: the result status is BEST_AVAILABLE_BELOW_THRESHOLD, the requested Minimum-Attendee Threshold remains 3, every ranked option has actual attendance 2, and the warning communicates requested 3 versus actual 2. BB-THR-03 proves that a best attendance below 2 returns NO_TIMESLOT_MEETS_THRESHOLD and performs no routing.

4 White-box Method 1: Interval Validation

Selected method: **AvailabilityService._validate_intervals()**. It sorts submissions, enforces the Session date range, rejects overlap, and returns a deterministic order. These branches protect availability data before persistence and recommendation generation.

4.1 Production source excerpt

```
138      def _validate_intervals(item: SchedulingSession, intervals: list[AvailabilityIntervalInput]):
139          ordered = sorted(intervals, key=lambda value: value.start_date_time)
140          for interval in ordered:
141              if _as_utc(interval.start_date_time) < _as_utc(item.date_range_start) or _as_utc(interval.end_date_time) > _as_utc(item.date_range_end):
142                  raise HTTPException(422, "Availability must be inside the session date range")
143          for previous, current in zip(ordered, ordered[1:]):
144              if current.start_date_time < previous.end_date_time: raise HTTPException(422, "Availability intervals cannot overlap")
145          return ordered
```

4.2 Control-flow graph

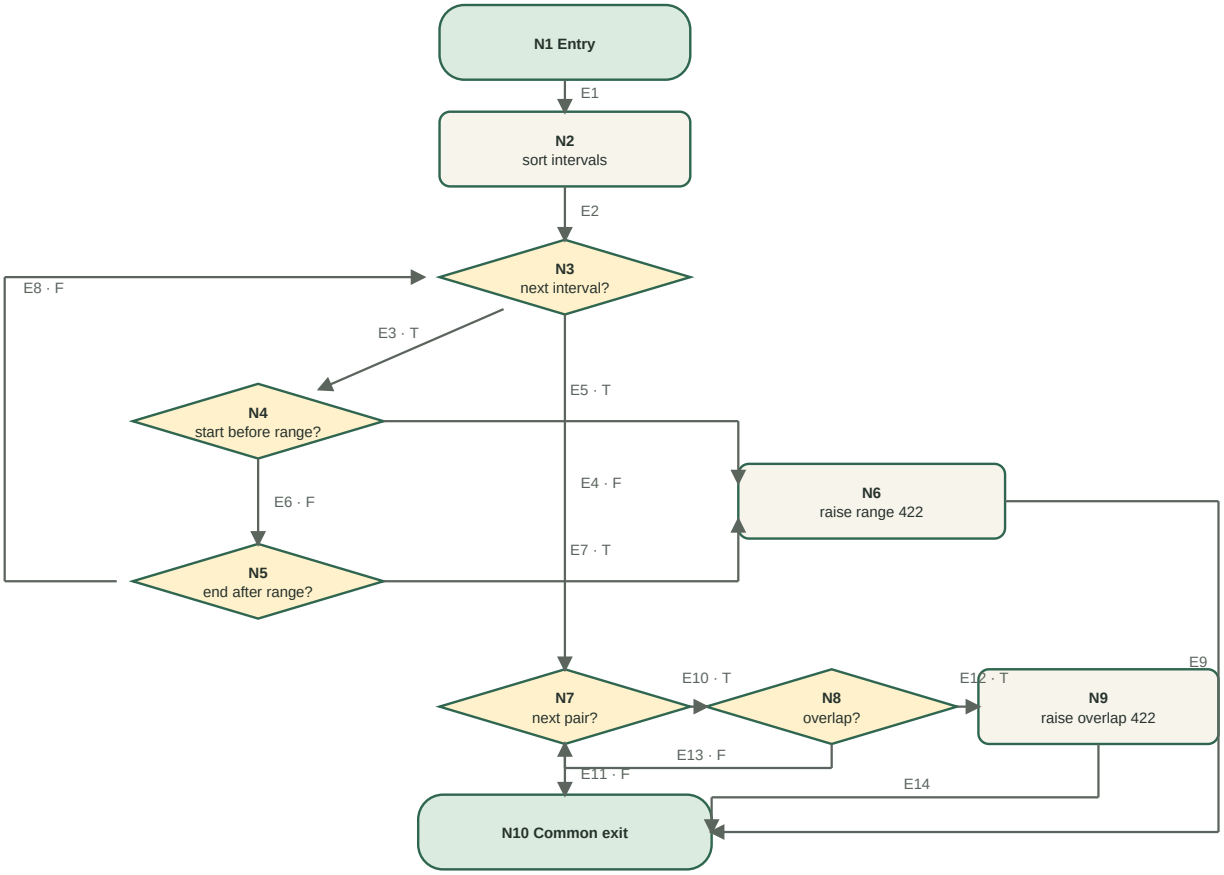

Figure 1. Fixed CFG convention: loop conditions and both range comparisons are decisions; exceptions converge at a common graph exit.

4.3 Cyclomatic complexity

Measure	Value
Nodes	N = 10
Edges	E = 14
Graph formula	14 - 10 + 2 = 6
Predicate formula	5 decisions + 1 = 6
Verified path-vector rank	6 over the rational numbers

4.4 Interval-validation Basis Paths

Case	Independent node path	Input	Expected	Actual	Result
WB-IV-01	N1-N2-N3(F)-N7(F)-N10	no intervals	[]	[]	PASS
WB-IV-02	N1-N2-N3(T)-N4(T)-N6-N10	start = range start - 1 min	HTTP 422 range rejection	HTTP 422: Availability must be inside the session date range	PASS
WB-IV-03	N1-N2-N3(T)-N4(F)-N5(T)-N6-N10	end = range end + 1 min	HTTP 422 range rejection	HTTP 422: Availability must be inside the session date range	PASS
WB-IV-04	N1-N2-N3(T)-N4(F)-N5(F)-N3(F)-N7(F)-N10	exact range endpoints	one unchanged interval	[(10:00–12:00)] SGT	PASS
WB-IV-05	N1-N2-N3(T)-N4(F)-N5(F)-N3(T)-N4(F)-N5(F)-N3(F)-N7(T)-N8(T)-N9-N10	two intervals overlap 1 min	HTTP 422 overlap rejection	HTTP 422: Availability intervals cannot overlap	PASS
WB-IV-06	N1-N2-N3(T)-N4(F)-N5(F)-N3(T)-N4(F)-N5(F)-N3(F)-N7(T)-N8(F)-N7(F)-N10	unsorted touching intervals	sorted; touching accepted	[(10:00–11:00), (11:00–12:00)] SGT	PASS

The six paths cover all 14 directed CFG edges. Their edge-incidence vectors have rational rank 6, so the path set is independent under the stated graph convention. IV-04 and IV-06 distinguish inclusive range endpoints and non-overlapping touching intervals from the two error paths.

Faults detected by this set

A missing range check fails IV-02 or IV-03; changing inclusive endpoints to strict ones fails IV-04; changing the overlap comparison from < to <= fails IV-06; omitting sorting fails the literal ordering assertion in IV-06; and returning an incorrect public status/detail fails IV-02, IV-03 or IV-05.

5 White-box Method 2: Participant Attendance

Selected method: _**ParticipantAvailability.status_for()**. It preserves NO_RESPONSE and WITHDRAWN, gives BUSY precedence over AVAILABLE, and requires full slot coverage. For CFG purposes each any(...) generator is expanded into iterator-next plus its two short-circuit comparisons; this convention is used consistently in graph, formulas and paths.

5.1 Production source excerpt

```
111      def status_for(self, start: datetime, end: datetime) -> AttendanceStatus:
112          p = self.participant
113          if p.record_status is None:
114              return AttendanceStatus.NO_RESPONSE
115          if p.record_status is RecordStatus.WITHDRAWN:
116              return AttendanceStatus.WITHDRAWN
117          # A BUSY block (e.g. imported from Google Calendar, FR-B8) that overlaps
118          # any part of the slot makes the Participant unavailable for it.
119          if any(bs < end and start < be for bs, be in self.busy):
120              return AttendanceStatus.UNAVAILABLE
121          # The slot must lie wholly inside one merged AVAILABLE block.
122          if any(as_ <= start and end <= ae for as_, ae in self.available):
123              return AttendanceStatus.AVAILABLE
124          return AttendanceStatus.UNAVAILABLE
```

5.2 Control-flow graph

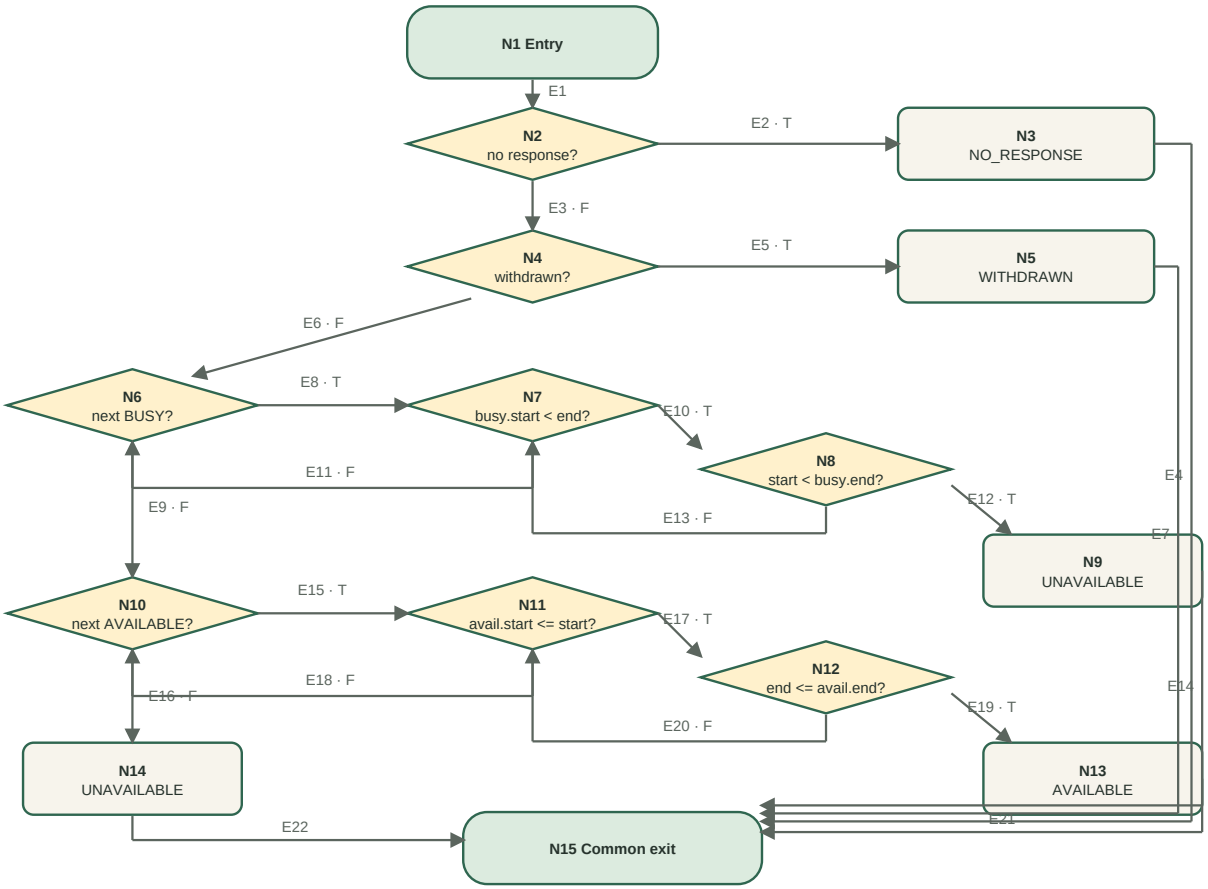

Figure 2. Expanded short-circuit CFG. Boundary paths show that intervals touching—but not crossing—the slot boundary do not overlap.

5.3 Cyclomatic complexity

Measure	Value
Nodes	N = 15
Edges	E = 22
Graph formula	22 - 15 + 2 = 9
Predicate formula	8 decisions + 1 = 9
Verified path-vector rank	9 over the rational numbers

5.4 Participant-attendance Basis Paths

Case	Independent node path	Input	Expected	Actual	Result
WB-PA-01	N1-N2(F)-N4(F)-N6(F)-N10(F)-N14-N15	active; no blocks	UNAVAILABLE	UNAVAILABLE	PASS
WB-PA-02	N1-N2(T)-N3-N15	record missing	NO_RESPONSE	NO_RESPONSE	PASS
WB-PA-03	N1-N2(F)-N4(T)-N5-N15	withdrawn; AVAILABLE exists	WITHDRAWN	WITHDRAWN	PASS
WB-PA-04	N1-N2(F)-N4(F)-N6(T)-N7(F)-N6(F)-N10(F)-N14-N15	BUSY starts at slot end	UNAVAILABLE; no overlap	UNAVAILABLE	PASS
WB-PA-05	N1-N2(F)-N4(F)-N6(T)-N7(T)-N8(F)-N6(F)-N10(F)-N14-N15	BUSY ends at slot start	UNAVAILABLE; no overlap	UNAVAILABLE	PASS
WB-PA-06	N1-N2(F)-N4(F)-N6(T)-N7(T)-N8(F)-N6(T)-N7(T)-N8(T)-N9-N15	earlier nonmatch; later 1-min BUSY overlap	UNAVAILABLE	UNAVAILABLE	PASS
WB-PA-07	N1-N2(F)-N4(F)-N6(F)-N10(T)-N11(F)-N10(F)-N14-N15	AVAILABLE starts 1 min late	UNAVAILABLE	UNAVAILABLE	PASS
WB-PA-08	N1-N2(F)-N4(F)-N6(F)-N10(T)-N11(T)-N12(F)-N10(F)-N14-N15	AVAILABLE ends 1 min early	UNAVAILABLE	UNAVAILABLE	PASS
WB-PA-09	N1-N2(F)-N4(F)-N6(F)-N10(T)-N11(T)-N12(F)-N10(T)-N11(T)-N12(T)-N13-N15	earlier nonmatch; later exact full coverage	AVAILABLE	AVAILABLE	PASS

The nine paths cover all 22 directed edges and have rational rank 9. PA-04 and PA-05 exercise opposite short-circuit boundary outcomes; PA-06 skips an earlier nonmatching BUSY block before a later overlap; PA-07 and PA-08 isolate the two full-containment comparisons; PA-09 skips an earlier nonmatching AVAILABLE block before a later full-cover match.

6 Supplemental Pre-routing Regressions

Seven existing pre_routing_exclusion() paths remain executable regression evidence, but they are not counted as either of the two assessed white-box methods. This prevents the report from discarding useful tests while keeping the Manual 3.2.2 scope precise.

Case	Regression condition	Observed	Result
WB-PR-01	below Session threshold	BELOW_THRESHOLD	PASS
WB-PR-02	park skips sports constraints	None	PASS
WB-PR-03	sports group-size failure	GROUP_SIZE	PASS
WB-PR-04	sports: no group/lead constraints	None	PASS
WB-PR-05	sports: meets group; no lead	None	PASS
WB-PR-06	booking lead fails by 1 hour	BOOKING_LEAD_TIME	PASS
WB-PR-07	exact booking-lead boundary	None	PASS

6.1 Production-configuration guard

Case	Observed relationship	Actual	Result
CFG-STEP-01	successive candidate starts	15-minute delta	PASS

6.2 Why supplemental

The assessed complexity and independent-path claims apply only to _validate_intervals() and status_for(). The seven WB-PR cases and one configuration guard are reported separately so they cannot inflate the 15 assessed white-box count.

Evidence group	Count	Pass	Assessment role
Black-box (BB)	25	25	Assessed
Interval paths (WB-IV)	6	6	Assessed method 1
Attendance paths (WB-PA)	9	9	Assessed method 2
Pre-routing regressions (WB-PR)	7	7	Supplemental
Production-step guard	1	1	Supplemental
Total	48	48	40 assessed + 8 supplemental

7 Execution Evidence and Cross-check

Verification	Observed result	Status
Black-box test file	25 passed	PASS
Assessed white-box file	15 passed	PASS
Supplemental pre-routing file	7 passed	PASS
Production-step guard	15-minute consecutive-start delta	PASS
Fresh Lab 4 JUnit	48 passed; 0 failed; 0 errors; 0 skips	PASS
Full backend regression	312 passed; one third-party deprecation warning	PASS
Frontend Node + UI tests	21 passed	PASS
Frontend lint	exit code 0	PASS
Frontend production build	152 modules transformed; exit code 0	PASS

7.1 One-to-one evidence mapping

The generator refuses to build unless the JUnit names are exactly the 40 assessment mappings plus eight supplemental mappings. Duplicate assessment IDs, missing cases, extra cases, failures, errors and skips are structural failures—not prose warnings.

Cross-check	Verified value
Assessment IDs	40 unique: 25 BB + 6 WB-IV + 9 WB-PA
Supplemental IDs	7 unique WB-PR + 1 production-step guard
JUnit total	48 unique passing test names
Interval graph	N=10, E=14, full edge coverage, path rank 6
Attendance graph	N=15, E=22, full edge coverage, path rank 9
SRS oracle	v2.3 contains both recovery statuses and requested-versus-actual disclosure
Secrets/placeholders	Rejected by final extracted-text assertions

8 SRS v2.3 Traceability and Consistency

Requirement / rule	Executed evidence	Consistency result
FR-A1.1 duration 15-480	BB-DUR-01..06	Invalid neighbours rejected; endpoints and valid neighbours accepted
FR-A4 positive threshold	BB-THR-04..05	0 rejected; explicit 1 accepted
FR-D1 normal threshold	BB-THR-01	Threshold-met recommendations remain normal
FR-D1.1 recovery	BB-THR-02..03	Near match labelled; below system minimum returns no candidates/routing
FR-E1.1 disclosure	BB-THR-02	Requested 3 preserved; actual 2; warning contains both values
FR-C3 activity support	BB-VEN-01..02	Unsupported venues excluded before shortlist/ranking
FR-D3 60-minute cap	BB-JRN-01..02	60 retained; 61 excluded
NFR-3 missing data	BB-RTE-01..03	No zero duration fabricated; unavailable/error remains explicit
Availability interval integrity	WB-IV-01..06	Range, overlap, sorting and touching boundaries verified
Attendance-state semantics	WB-PA-01..09	Four states, BUSY precedence and full coverage verified

8.1 Pilot near-match decision

The production Pilot intentionally avoids a dead end when the requested threshold is narrowly missed. If the best actual attendance is at least the default minimum of 2, it returns only the best-attended slots as BEST_AVAILABLE_BELOW_THRESHOLD, keeps the requested value unchanged, and warns with requested versus actual attendance. If best attendance is below 2, it returns NO_TIMESLOT_MEETS_THRESHOLD before routing. This is now stated in SRS v2.3 and tested without changing production code.

8.2 Internal consistency

The same fixed graph convention, node labels, edge sets, path IDs and complexity values are used by the DOT files, automated test names, generator data and this PDF. The report therefore has no conflict between diagrams, formulas and executed evidence.

9 Limitations, Conclusion and Reproduction

9.1 Limitations

The black-box control tests use a deterministic router at the external provider port so results do not depend on OneMap availability. They validate the application's handling of provider success, absence, malformed success and exceptions, but do not replace deployed integration smoke tests for Google OAuth, OneMap, PostgreSQL or Render. Basis-path coverage proves the selected control-flow paths, not every possible data combination or system-level concurrency condition.

9.2 Conclusion

The deliverable meets Lab 4 Manual sections 3.2.1-3.2.4 with current Pilot evidence: 25 black-box equivalence/boundary cases, 15 assessed basis paths for two class methods, and eight clearly separated supplemental guards. All 48 tests passed, the two CFG calculations and path-vector ranks agree, and the SRS v2.3 near-match behavior is fully traceable.

9.3 Reproduction

```
cd backend
python -m pytest tests/lab4 -q --junitxml=../docs/lab4/evidence/lab4-junit.xml
cd ..
python docs/lab4/generate_report.py
python docs/srs/update_v2_3.py 'docs/SRS Group Scheduling Application v2.3.docx'
```

Artifacts: docs/SRS Group Scheduling Application v2.3.docx/.pdf and docs/lab4/Lab4_Software_Testing_Report.pdf. Durable logs and JUnit are in docs/lab4/evidence at commit 57ee158.

Reference: SC2006 Lab 4 Manual, sections 3.2.1-3.2.4.